

# Desain Lengkap — Aplikasi Manajemen Internal

CV Ragafi Digital Technology

Tanggal: 27 November 2025

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## Ringkasan Eksekutif

Aplikasi web-based terpadu untuk mengelola seluruh kebutuhan internal CV Ragafi Digital Technology: HR & Payroll, Proyek & Task Management, Keuangan, Asset Management, Dokumen & SOP, serta modul Legal (NDA/SLA) dan Sekutu Management. Aplikasi dirancang modular, scalable, dan aman untuk operasional perusahaan kecil-menengah (SME) di bidang jasa pembuatan website dan pengolahan data.

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## Tujuan & Ruang Lingkup

- Mengotomatiskan dokumen legal & HR (kontrak, slip gaji, rekap kehadiran, data karyawan).
  - Menyederhanakan manajemen proyek, pelacakan waktu, dan pengelolaan SLA.
  - Menyediakan laporan keuangan dasar (laba rugi, neraca, arus kas).
  - Menyimpan & mengelola asset (hardware & digital), SOP, dan template dokumen.
  - Menyediakan audit trail & kontrol akses per peran.
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## Arsitektur Tingkat Tinggi

- **Frontend:** Next.js (React) — responsive, SSR/SSG optional.
  - **Backend:** Node.js (NestJS/Express) atau Laravel (PHP) — RESTful API.
  - **Database:** PostgreSQL atau MySQL/MariaDB.
  - **Storage:** S3 (AWS) atau MinIO.
  - **Auth:** JWT + Role-based Access Control (RBAC).
  - **Deployment:** Docker / Docker Compose, CI/CD via GitHub Actions. Nginx + Certbot untuk TLS.
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## User Roles & Permission (RBAC)

- **Super Admin (Pemilik / Sekutu)** — akses penuh, manage company settings.
- **Managing Partner (Direktur Utama)** — approve kontrak, keuangan, laporan.
- **CTO** — akses technical project, deployments, repos.
- **COO** — akses operasional, timeline, SLA.
- **HR Admin** — manage karyawan, payroll, attendance.
- **Finance** — invoice, cashbook, laporan keuangan.
- **Project Manager** — buat project, assign tasks.

- **Employee / Staff** — timesheet, view kontrak, dokumen SOP.
- **Client (Portal Opsional)** — lihat invoice, upload assets, ticket.

Kontrol akses granular: CRUD per resource + izin approve/sign.

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## Modul & Fitur Detail

### 1. HR & Payroll

- Employee Master Data (KTP, NPWP, BPJS, bank, emergency contact).
- Contract Management (generate PDF dari template + tanda tangan digital sederhana).
- Attendance (manual check-in/out, import CSV, integration device).
- Leave & Permit workflow (submit → approve/reject).
- Payroll Engine: komponen gaji pokok, tunjangan, lembur, potongan → slip gaji PDF.
- Performance Review & KPI tracking.
- Document Repository untuk KTP, kontrak, sertifikat.

### 2. Project & Task Management

- Project creation (scope, client, owner/PM, budget, timeline, SLA).
- Milestone & tasks (kanban + list), subtasks, dependencies.
- Timesheet per task (manual/timer) + spent\_hours.
- Progress report otomatis (weekly/monthly export PDF).
- Issue/Bug Tracker + priority, status, SLA.
- Templates untuk jenis project (website, data processing, dsb).

### 3. Finance

- Invoice generator (nomor otomatis, include VAT if needed).
- Expense & Purchase request approval flow.
- Cashbook & bank reconciliation.
- Basic accounting reports: Laba Rugi (P&L), Neraca, Arus Kas.
- Integration point for payment gateway (Midtrans/Xendit) optional.
- Export CSV/Excel & PDF.

### 4. Asset Management

- Inventory list: hardware, software license, domain, hosting.
- QR Code per asset (printable).
- Assignment history (who pakai), maintenance schedule.
- Depreciation tracking (optional).

### 5. Documents & Legal

- SOP Library (searchable, versioned, markdown/html preview).
- Template generation: NDA, SLA, Kontrak Kerja, Surat Penawaran.
- E-signature minimal (user approval with timestamp) or integrasi pihak ketiga.

- Document access rules & audit trail.

## 6. Communication & Notifications

- Internal announcements / surat edaran.
- Ticketing system untuk client support.
- In-app notifications + email triggers (invoice due, approval needed, deadline).
- Comment threads di tiap project/task.

## 7. Sekutu & Governance

- Manage sekutu (data, modal disetor, persentase kepemilikan).
- Pembagian laba otomatis berdasarkan aturan tersimpan.
- Dokumen rapat sekutu (MoM), perubahan perjanjian, versi akta.

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## Data Model — Tabel Inti (ERD Ringkas)

Tabel inti dengan field penting; gunakan tipe sesuai Postgres/MySQL.

**users** - id PK, name, email(unique), password\_hash, role\_id FK, employee\_id FK NULL, created\_at, updated\_at

**roles** - id PK, name, permissions JSON

**employees** - id PK, user\_id FK, nik, npwp, bpjs\_ketenagakerjaan, bpjs\_kesehatan, join\_date, position, bank\_name, bank\_account, salary\_base, allowances JSON, contract\_id FK, status

**contracts** - id, employee\_id FK, start\_date, end\_date, contract\_type, file\_url

**attendance** - id, employee\_id FK, date, checkin\_time, checkout\_time, work\_hours, note

**leave\_requests** - id, employee\_id, type, from\_date, to\_date, status, approver\_id

**projects** - id, code, name, client\_id, owner\_id, start\_date, end\_date, budget, status

**tasks** - id, project\_id, title, description, assignee\_id, status, priority, estimate\_hours, spent\_hours

**timesheets** - id, task\_id, employee\_id, date, hours, description

**invoices** - id, invoice\_no, client\_id, project\_id, amount, tax, due\_date, status, file\_url

**assets** - id, code, name, category, serial\_no, value, status, assigned\_to, purchase\_date

**sop\_documents** - id, title, content(markdown), file\_url, version, owner\_id

**audit\_logs** - id, user\_id, action, object\_type, object\_id, meta JSON, created\_at

(Catatan: full ERD dan SQL schema tersedia sebagai deliverable jika diminta.)

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## API Design (RESTful — contoh endpoint utama)

Semua endpoint protected (JWT) dan role-checked.

- POST /api/auth/login — {email, password} -> tokens
- GET /api/users/me
- GET /api/employees
- POST /api/employees
- GET /api/attendance?employee\_id=&month=
- POST /api/attendance/checkin — {employee\_id, timestamp}
- POST /api/projects
- GET /api/projects/:id
- POST /api/projects/:id/tasks
- POST /api/tasks/:id/timesheet
- POST /api/invoices
- GET /api/reports/pnl?from=&to=
- GET /api/assets
- POST /api/documents/generate/contract — generate PDF from template
- POST /api/notifications/send

(OpenAPI / Swagger spec dapat disertakan sebagai deliverable.)

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## UI / UX — Layar Utama & Wireframes (Deskripsi)

1. **Dashboard:** widget KPIs (projects active, invoice unpaid, payroll total), activity feed.
2. **Employee Directory:** tabel filterable, quick actions (generate kontrak, lihat dokumen).
3. **Payroll:** pilih periode → kalkulasi gaji → generate slip (batch).
4. **Attendance:** calendar & rekap per karyawan.
5. **Projects (Kanban):** drag & drop, card details.
6. **Project Detail:** tabs (Overview / Tasks / Files / Timesheet / Reports / SLA).
7. **Finance:** list invoices, create invoice, mark paid, export.
8. **Assets:** list, QR code, assign/unassign.
9. **Documents:** SOP viewer, template generator.
10. **Settings:** roles, company profile, bank, tax settings.

Design guideline: clean, minimal, responsive. Use component library + Tailwind.

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## Security & Compliance

- Password hashing (bcrypt/argon2), HTTPS/TLS mandatory.
  - RBAC + per-record ACL untuk dokumen sensitif.
  - PII encryption at rest (NPWP, bank account).
  - Audit logging for financial & legal actions.
  - Backups daily ke S3, retention minimal 30 hari.
  - Optional 2FA untuk sekutu & finance.
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## Integrasi & Eksternal

- Payment gateway: Midtrans / Xendit / QRIS.
  - Email: SendGrid / Mailgun / SMTP.
  - GitHub/GitLab webhook untuk deployment status.
  - Accounting export (CSV) for local accountant.
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## Deployment & Infrastruktur (Rekomendasi)

- Dockerized apps (frontend & backend).
  - Managed DB (DigitalOcean Managed Postgres / AWS RDS) atau self-hosted MariaDB.
  - Object storage: S3 / MinIO.
  - Reverse proxy + SSL: Nginx + Certbot.
  - CI/CD: GitHub Actions (build -> test -> deploy).
  - Monitoring: Sentry (errors) + Prometheus/Grafana (metrics).
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## QA, Testing & Monitoring

- Unit tests + integration tests.
  - E2E (Playwright/Cypress) untuk critical flows (payroll, invoice, contract generation).
  - Load testing for reports (k6).
  - Security scan (OWASP ZAP) sebelum production.
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## Phased Delivery Plan (MVP → Full)

**Phase 1 — MVP (8–12 minggu)** - Auth & RBAC, Employee master, Project basic, Tasks (kanban), Attendance basic, Document upload, Basic Invoice, Dashboard.

**Phase 2 — Payroll & Finance (6–8 minggu)** - Payroll engine, batch slip generation, expense module, P&L report.

**Phase 3 — Assets, SOP & Legal (4–6 minggu)** - Asset management, SOP library, template generator, NDA/SLA flows, e-sign integration.

**Phase 4 — Integrasi & Automasi (ongoing)** - Payment gateway, email automation, backups, monitoring, advanced reports, client portal.

(total estimasi scope full product: 4–6 bulan untuk core + fitur matang)

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## Deliverables yang bisa saya siapkan sekarang

- Dokumen desain lengkap (DOCX / PDF) — berisi semua bagian di atas + ERD visual.
- SQL schema (CREATE TABLE) untuk semua tabel.
- OpenAPI (Swagger) stub / Postman collection.
- UI mockups (6 layar utama) dan flowchart.

Sebutkan file mana yang kamu mau, dan saya akan generate untukmu.

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## Catatan Tambahan

- Semua template dokumen (kontrak, NDA, SLA, surat penawaran) disiapkan dengan placeholder yang mudah di-fill.
  - Sistem dapat dihost di VPS internal jika ingin kontrol penuh, tapi managed services direkomendasikan untuk reliability.
  - Untuk ekonomi pengembangan: dapat mulai dengan 1 full-stack dev + 1 backend + 1 UI/UX (kontrak 3–6 bulan) atau outsourcing ke agency.
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## Akhir Dokumen

Jika kamu mau, aku akan: - Generate **DOCX + PDF** dari desain lengkap ini. - Buat **SQL schema** dan **Postman / OpenAPI** stub. - Buat **UI mockups** (PNG) untuk 6 layar utama.

Tulis file mana yang kamu mau aku buat sekarang:  ,  ,  ,  , .